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§3.3 The minimal polynomial

<u>Motivation</u>: $A \in M_n$, char. poly.

$$p_A(t) = \det(tI - A) \quad (\deg p_A(t) = n)$$

& Cayley-Hamilton $\Rightarrow$ $p_A(A) = 0$

Is there another polynomial that annihiliates A and is of lesser degree $(\leq n)$?

<u>Definition</u>: Given $A \in M_n$, there exists a monic polynomial $q_A(t)$ of minimum degree $(\leq n)$ that annihiliates A, i.e. $q_A(A) = 0$. We call $q_A(t)$ the <u>minimal polynomial</u> of A

<u>Facts</u>: (1) $q_A(t)$ exists (since $p_A(A) = 0$)

(2) $q_A(t) \mid p_A(t)$ (minimal divides characteristic!) $\xrightarrow{\text{proof}}$

Since $\deg q_A(t) \leq \deg p_A(t)$, there exist polynomials $h(t)$ & $r(t)$ such that

$$p_A(t) = q_A(t)\,h(t) + r(t)$$

$$\underset{\substack{\uparrow \\ \text{remainder}}}{\deg r(t)} < \deg q_A(t)$$

But then

$$0 = p_A(A) = \overset{=0}{\cancel{q_A(A)}}\,h(A) + r(A)$$

$$= r(A)$$

So $r(A) = 0$. By minimality of degree of $q_A(t)$, it follows $r(t) \equiv 0$ $(r(t) = 0, \forall t)$

That is,

$$p_A(t) = q_A(t)\,h(t)$$

i.e.

$$q_A(t) \mid p_A(t).$$

(i.e. $q_A(t)$ is a factor of $p_A(t)$)

(Note: this also proves uniqueness of minimal polynomial!)

(3) Similar matrices have the same minimal polynomials:

<u>Proof:</u> $A = SBS^{-1}$ $(A \sim B)$. Then

$$q_B(A) = q_B(SBS^{-1})$$
$$= \cdots$$
$$= S\, q_B(B)\, S^{-1}$$
$$= S \cdot 0 \cdot S^{-1} = 0$$

i.e. $q_B(t)$ annihilates A

Thus $\deg q_B(t) \geq \deg q_A(t)$

Similarly $\deg q_A(t) \geq \deg q_B(t)$

i.e. $\left.\begin{array}{l} \deg q_B(t) = \deg q_A(t) \\ q_B(A) = q_A(t) \end{array}\right\}$

$\Rightarrow$ $q_B(t) = q_A(t)$ by uniqueness of $q_A(t)$!

(4) $q_A(\lambda) = 0 \iff \lambda \in \sigma(A)$

<u>Proof</u> $(\Rightarrow)$ $p_A(t) = q_A(t)\, h(t)$ as above, so

if $q_A(\lambda) = 0$, then $p_A(\lambda) = 0$,

i.e. $\lambda \in \sigma(A)$.

($\Leftarrow$) If $\lambda \in \sigma(A)$: $\exists\, x \neq 0$ such that $Ax = \lambda x$.

But then

$$0 = 0x = q_A(A)x \overset{Ax=\lambda x}{=} q_A(\lambda)x, \quad x \neq 0$$

$$\Rightarrow \quad q_A(\lambda) = 0 \quad \blacksquare$$

Intimate connection of $q_A(t)$, $p_A(t)$ & JCF(A)

(that allows us to explicitly find $q_A(t)$)

Suppose $A = J_n(\lambda) = J \in M_{n,n}$. Clearly

$$p_A(t) = p_J(t) = (t-\lambda)^n$$

& $J - \lambda I = J_n(0)$ so

$$(J-\lambda I)^k \neq 0 \quad \forall k < n$$

$$(J-\lambda I)^n = 0$$

Since $q_A(t) \mid p_A(t)$, i.e. $q_A(t) = (J-\lambda I)^k$

we'll have to have $k = n$, i.e.

$$q_A(t) = (t-\lambda)^n$$

Lift this analysis up to a general Jordan matrix; gives:

<u>Fact</u> For a general Jordan matrix J, $q_J(t)$ must contain a factor

$$(t-\lambda_i)^{v_i}$$

for each distinct eigenvalue $\lambda_i \in \sigma(J)$, where v_i = order of the largest Jordan block corresponding to λ_i

$$= \underline{\text{index}}(\lambda_i)$$

<u>Example</u>

$$A = J = J_2(1) \oplus J_3(0) \oplus J_2(0) \oplus J_1(1) \oplus J_3(5)$$

$$\sigma(A) = \{1,1,1,0,0,0,0,0,5,5,5\}$$

$$p_A(t) = (t-1)^3 (t^5)(t-5)^3$$

$$q_A(t) = (t-1)^2 (t^3)(t-5)^3$$

$$= t^3 (t-1)^2 (t-5)^3$$

$$q_A(t) \mid p_A(t), \quad q_A(A) = 0$$

index$(0) = 3$, index$(1) = 2$, index$(5) = 3$

Consequence:

Theorem Let $A \in M_n$. The following are equivalent:

(a) A is diagonalizable

(b) $q_A(t)$ comprises only distinct linear factors

(c) Every root of $q_A(t)$ has multiplicity 1

(d) $\forall \hat{t} : q_A(\hat{t}) = 0 \Rightarrow q_A'(\hat{t}) \neq 0$

Next time:

We say for A having JCF with only one block,

$$q_A(t) \equiv p_A(t).$$

Characterize <u>all</u> matrices A such that minimal $\equiv$ characteristic.

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<u>Question</u> Given $p(t) = t^n + a_{n-1}t^{n-1} + \cdots + a_1 t + a_0$ (any monic polynomial), is there a matrix $A \in M_n$ such that $q_A(t) = p(t)$ $(= p_A(t))$?

<u>Answer</u>: Yes, e.g.

$$A = \begin{bmatrix} 0 & 0 & \cdots & 0 & -a_0 \\ 1 & 1 & & \vdots & \vdots \\ & & \ddots & 0 & \vdots \\ 0 & & & 0 & -a_{n-2} \\ & & & 1 & -a_{n-1} \end{bmatrix}$$

e.g. $p(t) = t^3 - 2t^2 + 3t + 4$

$$A = \begin{bmatrix} 0 & 0 & -4 \\ 1 & 0 & -3 \\ 0 & 1 & 2 \end{bmatrix}, \quad q_A(t) = p(t) = p_A(t)$$

Justification $\longrightarrow$

<u>Given</u> $p(t) = t^n + a_{n-1}t^{n-1} + \dots + a_1 t + a_0$

$$A = \begin{bmatrix} 0 & & & -a_0 \\ 1 & \ddots & & \vdots \\ & \ddots & 0 & \\ & & 1 & -a_{n-1} \end{bmatrix}$$

(e_k = k-th column of identity I_n)

we have:

$$e_1 = A^0 e_1$$

$$Ae_1 = e_2 = Ae_1$$

$$Ae_2 = e_3 = A^2 e_1$$

$$\vdots$$

$$Ae_{n-1} = e_n = A^{n-1} e_1$$

So

$$Ae_n = -a_{n-1}e_n - a_{n-2}e_{n-1} - \dots - a_1 e_2 - a_0 e_1$$

$$= -a_{n-1}A^{n-1}e_1 - a_{n-2}A^{n-2}e_1 - \dots - a_1 Ae_1 - a_0 e_1$$

$$= (A^n - p(A))e_1 \xRightarrow{A^n e_1 = Ae_n} \boxed{p(A)e_1 = 0}$$

Hence

$$p(A)e_k = p(A)A^{k-1}e_1 = A^{k-1}p(A)e_1 = 0$$

$$k = 1, 2, \dots, n$$

Thus

$$p(A) = 0$$

$$\left(\begin{aligned} \cancel{Ae_n} &= \cancel{A^n e_1} - p(A)e_1 \\ &\text{but } Ae_n = A^n e_1 \\ &\Rightarrow -p(A)e_1 = 0 \\ &\Rightarrow \boxed{p(A)e_1 = 0} \end{aligned} \right)$$

So $p(t)$ is an annihilating polynomial of A of degree n & monic.

It's also of minimal degree because:

if $\begin{cases} q(t) = t^m + b_{m-1}t^{m-1} + \dots + b_1 t + b_0 \\ m < n, \quad q(A) = 0 \end{cases}$

then

$$0 = q(A) = A^m + b_{m-1}A^{m-1} + \dots + b_1 A + b_0 I$$

$$\Rightarrow 0 = q(A)e_1 = A^m e_1 + b_{m-1}A^{m-1}e_1 + \dots + b_1 Ae_1 + b_0 e_1$$

$$= e_{m+1} + b_{m-1}e_m + \dots + b_1 e_2 + b_0 e_1$$

$\Rightarrow \{e_1, e_2, \dots, e_{m+1}\}$ is linearly dependent

A contradiction, so no such $q(t)$ exists.

Thus $\boxed{p(t) = q_A(t) = p_A(t)}$

<u>Terminology</u> We call the A used above the <u>companion matrix</u> to the given polynomial $p(t)$

$$(p(t) = q_A(t) = p_A(t))$$

<u>Theorem</u> Every monic polynomial is both the characteristic and the minimal polynomial of its companion matrix

<u>Fact</u> Finding the eigenvalues of the companion matrix constitutes the "state of the art" method for finding the roots of any polynomial.
(Find/read : J. Auvenz & D. Watkins award winning paper)

<u>Fact</u> Companion matrices are notoriously ill-conditioned.

"$\text{cond}(A) = \|A\| \|A^{-1}\|$"

$$\text{cond}(C) \gg 1$$
$$\| \quad$$
$$\|C\| \|C^{-1}\|$$

Last thing about JCF

$A \in M_n$, $J = JCF(A)$, $S^{-1}AS = J$

What does S look like?

(J = diagonal $\iff$ S has n eigenvectors of A as its columns)

Let's look at $A = J_k(\lambda) = \begin{bmatrix} \lambda & 1 & & 0 \\ & \lambda & \ddots & \\ & & \ddots & 1 \\ 0 & & & \lambda \end{bmatrix}$

$\cancel{S = S J_k(\lambda)}$

Notice

$$J_k(\lambda) e_1 = \lambda e_1$$

$$J_k(\lambda) e_2 = \begin{bmatrix} 1 \\ \lambda \\ 0 \\ \vdots \\ 0 \end{bmatrix} = \lambda e_2 + e_1$$

$$J_k(\lambda) e_3 = \begin{bmatrix} 0 \\ 1 \\ \lambda \\ 0 \\ \vdots \\ 0 \end{bmatrix} = \lambda e_3 + e_2$$

$$\vdots$$

$$J_k(\lambda) e_{k-1} = \lambda e_{k-1} + e_{k-2}$$

$$J_k(\lambda) e_k = \lambda e_k + e_{k-1}$$

So $\{e_1, e_2, \dots, e_k\}$

are what we call a <u>Jordan chain</u>

of <u>generalized eigenvectors</u> of $J_k(\lambda)$

(e_1 = eigenvector, $e_2, \dots, e_k$ = "almost eigenvectors")

For general A, if $S = \{s_1, s_2, \dots, s_k\}$

& if $S^{-1}AS = J_k(\lambda)$

(since $AS = SJ_k(\lambda)$)

we have

$$\begin{cases} As_1 = \lambda s_1 \\ As_2 = \lambda s_2 + s_1 \\ \vdots \\ As_k = \lambda s_k + s_{k-1} \end{cases}$$

So S has a Jordan chain $\{s_1, s_2, \dots, s_k\}$

as its columns.

Generalizes to J = Jordan matrix, $A = SJS^{-1}$

The columns of S are collection of Jordan

chains corresponding to each Jordan block

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CHAPTER 4 : Hermitian & Symmetric Matrices

§ 4.1 Definitions, properties, characterizations

Definitions If $A = [a_{ij}] \in M_n$, we call A

- hermitian if $A = A^*$
- skew-hermitian if $A = -A^*$

(where $A^* = \overline{A^T}$)

Observations

(1) $\forall A \in M_n \Rightarrow A + A^*, AA^*, A^*A$ are all hermitian.

$(A+A^*)^* = A^* + A$, $(AA^*)^* = AA^*$ etc.

(2) If A is hermitian, then A^k hermitian for all $k \in \mathbb{N}$. Also, if A is invertible hermitian, then so is A^{-1}

$$(A^{-1})^* = (A^*)^{-1} = A^{-1}$$

(3) A, B hermitian $\Rightarrow \alpha A + \beta B$ is hermitian, $\forall \alpha, \beta \in \mathbb{R}$.

(same for skew-hermitian)

(4) $\forall A \in M_n \implies A - A^*$ is skew-hermitian

(5) $\forall A \in M_n$ we can write

$$A = \underbrace{\frac{A + A^*}{2}}_{\text{hermitian part of } A} + \underbrace{\frac{A - A^*}{2}}_{\text{skew-hermitian part of } A}$$

(6) Real matrices : $A^* = A^T$. So hermitian real matrices satisfy $A = A^T$; known as <u>symmetric</u>.

(7) Notice that the diagonal entries of every hermitian matrix are real!

$$\overline{a_{ii}} = a_{ii}, \quad i = 1, 2, \dots, n$$

<u>Theorem</u> Every $A \in M_n$ can be uniquely written as $A = S + iT$, where

S is hermitian

T is " hermitian

$i = \sqrt{-1}$

(resembles $z = \operatorname{Re} z + i \operatorname{Im} z \in \mathbb{C}$, $\operatorname{Re} z, \operatorname{Im} z \in \mathbb{R}$.
For matrices "hermitian" replaces "real")

<u>Proof</u> Note that

$$A = \frac{A+A^*}{2} + i\,\frac{A-A^*}{2i}$$

$S = \frac{A+A^*}{2}$, $T = \frac{A-A^*}{2i}$ are clearly hermitian

$$\left(T^* = \frac{A^*-A}{-2i} = \frac{A-A^*}{2i} = T \right)$$

Uniqueness:

If $A = E + iF$, $E = E^*$, $F = F^*$, then

$$2S = A + A^* = (E+iF) + (E+iF)^*$$

$$= E + \cancel{iF} + E^* - \cancel{iF^*} \qquad \begin{matrix}(F=F^*)\\(E=E^*)\end{matrix}$$

$$= 2E$$

$\Rightarrow S = E$

Similarly, $F = T$ ∎

The essense of hermitian matrices:

Theorem Let $A \in M_n$ be hermitian. Then

(a) $\forall x \in \mathbb{C}^n$, $x^* A x \in \mathbb{R}$

(b) $\sigma(A) \subseteq \mathbb{R}$

(c) $S^* A S$ is hermitian, $\forall S \in M_n$

Proof

(a)
$$\overline{x^* A x} = (x^* A x)^*$$
$$= x^* A^* (x^*)^*$$
$$\overset{A=A^*}{=} x^* A x \Rightarrow x^* A x \in \mathbb{R} \checkmark$$

(b) Say $\lambda \in \sigma(A)$: $Ax = \lambda x$, $x \neq 0$

Then $x^* A x = \lambda x^* x \quad (x^* x \neq 0)$

$$\Rightarrow \lambda = \frac{x^* A x}{x^* x} \in \mathbb{R} \checkmark$$

(real by (a) / positive)

(c) $(S^* A S)^* = S^* A^* S = S^* A S \checkmark$

■

It turns out we can state characterizations of $\longrightarrow$ hermitian matrices based on (a).

<u>Theorem</u> Let $A \in M_n$. The following are equivalent:

(i) A is hermitian

(ii) $x^* A x \in \mathbb{R}$ for all $\boxed{x \in \mathbb{C}^n}$

(iii) A is normal $(AA^* = A^*A)$ & $\sigma(A) \subseteq \mathbb{R}$

(iv) $S^* A S$ is hermitian for all $S \in M_n$

<u>Proof</u>

(i) $\Rightarrow$ (ii), (i) $\Rightarrow$ (iii) is by previous theorem.

[(ii) $\Rightarrow$ (i)] Note

$$(x+y)^* A (x+y) = x^* A x + y^* A y + x^* A y + y^* A x \in \mathbb{R}, \text{ for all } x, y \in \mathbb{C}^n \quad (\#)$$

Choose $x = e_k$, $y = e_j$. Then by $(\#)$,

$$(x+y)^* A (x+y) = \underbrace{a_{kk}}_{\in \mathbb{R}} + \underbrace{a_{jj}}_{\in \mathbb{R}} + a_{kj} + a_{jk} \in \mathbb{R}$$

$$\Rightarrow \quad a_{kj} + a_{jk} \in \mathbb{R}, \ \forall j, k.$$

$$\Rightarrow \quad \boxed{\operatorname{Im} a_{kj} = -\operatorname{Im} a_{jk}} \quad (1)$$

Next choose $x = ie_k$, $y = e_j$

As above, by (#)

$$-i a_{kj} + i a_{jk} \in \mathbb{R}$$

$$\Rightarrow \boxed{\operatorname{Re} a_{kj} = \operatorname{Re} a_{jk}} \quad (2)$$

By (1) & (2), $\overline{a_{kj}} = a_{jk}$, $\forall k, j$

i.e. $A = A^*$, i.e. (i) holds.

[(iii) $\Rightarrow$ (i)] A normal $\Rightarrow \exists$ unitary U : $A = U \Lambda U^*$ where Λ = diagonal on $M_n(\mathbb{R})$ since $\sigma(A) \subseteq \mathbb{R}$.

So $A^* = (U \Lambda U^*)^* = U \Lambda^* U^* = U \Lambda U^*$
$= A$.

[(iv) $\Rightarrow$ (i)] Choose $S = I$. ∎